

[illegible]

1. mPAP >20 mmHg

WHAT YOU SEE

Large gauge needle pointing to >20 mmHg

WHAT IT ENCODES

Pulmonary hypertension is now defined as a mean pulmonary arterial pressure (mPAP) >20 mmHg AT REST, measured invasively by right heart catheterization (normal mPAP is ~14 mmHg). mPAP is the integrated pressure, not the systolic PA pressure echo estimates. The 2022 ESC/ERS guidelines added a PVR threshold (>2 Wood units) to the definition so that PH also requires evidence of true pulmonary vascular disease, not just high flow or high left-sided pressure.

BOARD TRAP

WRONG: defining PH only by an elevated PA SYSTOLIC pressure on echo. Discriminator: the definition uses MEAN PAP >20 mmHg by right heart catheterization — echo only screens/estimates and can over- or under-call PH.

2. >25 is the OLD rule

WHAT YOU SEE

Second gauge reading >25 with a red X over it

WHAT IT ENCODES

The legacy threshold of mPAP ≥25 mmHg is OUTDATED. The cutoff was lowered to >20 mmHg because the 20-24 mmHg 'borderline' range carries increased mortality and disease progression, allowing EARLIER detection. Note PH is still NOT diagnosed on the borderline range alone without an abnormal PVR.

BOARD TRAP

WRONG: choosing mPAP ≥25 mmHg as the current definition of PH. Discriminator: ≥25 is the historical number — the modern hemodynamic cutoff is mPAP >20 mmHg (with PVR >2 WU for pre-capillary disease).

3. Abnormal PVR >2 WU

WHAT YOU SEE

Small gauge marked >2 Wood units, labeled abnormal PVR

WHAT IT ENCODES

Pulmonary vascular resistance (PVR) = (mPAP – PAWP) / cardiac output, expressed in Wood units. PVR >2 WU defines abnormal PRE-CAPILLARY vascular remodeling and is now part of the PAH definition. It distinguishes a vessel/remodeling problem from a pure flow or pressure-transmission problem.

BOARD TRAP

WRONG: assuming any elevated mPAP means high PVR / intrinsic vascular disease. Discriminator: high mPAP with NORMAL PVR (≤2 WU) and high PAWP is post-capillary (left heart) disease, not a remodeled pulmonary bed — PVR is what proves the vasculature itself is diseased.

4. >3 WU outdated

WHAT YOU SEE

Plaque reading >3 WU with red X, labeled outdated

WHAT IT ENCODES

The prior PVR cutoff of >3 Wood units has been LOWERED to >2 WU in the 2022 criteria, paralleling the lower mPAP threshold to capture earlier pulmonary vascular disease.

BOARD TRAP

WRONG: using PVR >3 WU as the modern threshold for pre-capillary PH. Discriminator: >3 WU is the old number; current criteria use PVR >2 WU.

5. Pre-capillary pipe

WHAT YOU SEE

Blue pipe labeled PRECAPILLARY with PAWP ≤15 and PVR >2 tags

WHAT IT ENCODES

PRE-capillary PH: PAWP ≤15 mmHg AND PVR >2 WU (with mPAP >20). The problem is upstream of the pulmonary capillaries — the arterioles/vessels themselves. This profile defines WHO Groups 1 (PAH), 3 (lung disease/hypoxia), 4 (CTEPH), and 5 (multifactorial). Treat Group 1 with PAH-specific vasodilators; do NOT give those drugs to Groups 2/3 reflexively.

BOARD TRAP

WRONG: starting PAH-specific therapy (e.g., endothelin antagonists, PDE5 inhibitors, prostacyclins) for any pre-capillary picture. Discriminator: PAH-targeted drugs are validated for Group 1; in Group 3 they can worsen V/Q mismatch and hypoxemia — confirm the WHO group first.

6. Isolated post-capillary

WHAT YOU SEE

Red pipe labeled ISOLATED POSTCAPILLARY, PAWP >15, PVR ≤2

WHAT IT ENCODES

ISOLATED POST-capillary PH: PAWP >15 mmHg WITH normal PVR ≤2 WU. The pulmonary vasculature is passively transmitting elevated LEFT-heart pressure (HFrEF, HFpEF, valvular disease) — WHO Group 2, the MOST COMMON cause of PH overall. Treatment targets the left heart (diurese, treat HF/valve), not the pulmonary arteries.

BOARD TRAP

WRONG: treating Group 2 (left heart) PH with pulmonary vasodilators. Discriminator: isolated post-capillary PH (high PAWP, normal PVR) responds to optimizing left-heart disease; PAH drugs can cause pulmonary edema by increasing flow into a congested left atrium.

7. Combined pre+post

WHAT YOU SEE

Purple pipe labeled COMBINED, PAWP >15, PVR >2

WHAT IT ENCODES

COMBINED pre- and post-capillary PH (Cpc-PH): PAWP >15 mmHg AND PVR >2 WU. Long-standing left-heart disease causes secondary pulmonary vascular remodeling on top of the passive pressure rise — a worse prognosis than isolated post-capillary disease. Seen in Group 2 and contributing in Groups 4/5.

BOARD TRAP

WRONG: labeling combined pre/post-capillary PH as PAH and treating it like Group 1. Discriminator: the elevated PAWP keeps it rooted in left-heart disease (Group 2); the high PVR just marks added remodeling — manage the underlying left heart first.

8. PAWP = the knife

WHAT YOU SEE

Worker holding a knife labeled PAWP = THE KNIFE near gauge at 15

WHAT IT ENCODES

Pulmonary artery WEDGE pressure (PAWP, ≈ PCWP) is the single number that CUTS pre- from post-capillary PH. It estimates left atrial / left ventricular end-diastolic pressure. Cutoff: ≤15 mmHg = pre-capillary; >15 mmHg = post-capillary (left heart). Measure it at end-expiration and at euvolemia for accuracy.

BOARD TRAP

WRONG: ignoring PAWP and classifying PH by mPAP alone. Discriminator: without PAWP you cannot tell PAH from left-heart PH — the wedge pressure is the dividing line, and it changes management entirely.

9. PAWP <=15 = pre

WHAT YOU SEE

Tag: PAWP <=15 = precapillary

WHAT IT ENCODES

PAWP ≤15 mmHg points to PRE-capillary disease — the elevated pressure originates in the pulmonary vasculature itself (intrinsic arteriolar remodeling, hypoxic vasoconstriction, or thromboembolic obstruction), i.e., Groups 1, 3, 4, 5.

BOARD TRAP

WRONG: assuming a low/normal wedge automatically means Group 1 PAH. Discriminator: PAWP ≤15 only confirms PRE-capillary — you still must distinguish Group 1 vs Group 3 (lung disease) vs Group 4 (CTEPH) by PFTs, imaging, and V/Q scan.

10. PAWP >15 = left heart

WHAT YOU SEE

Tag: PAWP >15 = left heart

WHAT IT ENCODES

PAWP >15 mmHg indicates LEFT HEART disease (WHO Group 2) — HFrEF, HFpEF, or left-sided valvular disease — the MOST COMMON cause of PH worldwide. The high left atrial pressure transmits backward into the pulmonary circulation.

BOARD TRAP

WRONG: calling an elevated PAWP a Group 1 PAH finding. Discriminator: PAH (Group 1) requires a LOW wedge (≤15); a high PAWP shifts the diagnosis to Group 2 left-heart disease and away from PAH-specific therapy.

11. Group tags by pipe

WHAT YOU SEE

Color board listing PVR/PAWP combinations mapped to Groups 1/3/4/5, Group 2, Groups 2/4/5

WHAT IT ENCODES

The hemodynamic profile maps onto the WHO 5-group classification: PRE-capillary (PAWP ≤15, PVR >2) = Groups 1 (PAH: idiopathic, heritable, drug/toxin, connective-tissue disease, congenital shunts), 3 (lung disease/hypoxia), 4 (CTEPH), 5 (multifactorial). ISOLATED POST-capillary (PAWP >15, PVR ≤2) = Group 2 (left heart). COMBINED (PAWP >15, PVR >2) = Group 2 with remodeling, and seen in 4/5.

BOARD TRAP

WRONG: equating 'pulmonary hypertension' with PAH (Group 1). Discriminator: Group 1 PAH is the rare pre-capillary subset; Group 2 (left heart) and Group 3 (lung disease) are far more common — the WHO group, not just the pressures, drives treatment.

12. Diuresis trap

WHAT YOU SEE

Red warning sign DIURESIS TRAP, post-capillary now looks pre-capillary, masked gauge

WHAT IT ENCODES

Aggressive DIURESIS lowers left-heart filling pressure, so a true Group 2 (post-capillary) patient can be catheterized with a falsely normal PAWP (≤15) and be MISLABELED as pre-capillary PAH. Always measure hemodynamics at EUVOLEMIA; if the wedge is borderline, a fluid challenge or exercise can unmask occult left-heart disease (HFpEF).

BOARD TRAP

WRONG: diagnosing PAH (Group 1) after a recently/over-diuresed patient shows a normal wedge. Discriminator: catheterize at euvolemia and use a provocative fluid challenge — a wedge that rises with volume reveals masked left-heart (Group 2) disease, not PAH.

13. COPD barrel chest

WHAT YOU SEE

Barrel-chested COPD figure with mPAP 44, PAWP 19, PVR >2 tags

WHAT IT ENCODES

The barrel chest signals WHO Group 3 (PH due to lung disease / chronic hypoxia, classically COPD or ILD) — chronic hypoxic pulmonary vasoconstriction remodels the vessels. When a COPD patient ALSO has left-heart disease (high PAWP) plus high PVR, the result is COMBINED PH (here mPAP 44, PAWP 19, PVR >2). Group 3 PH is usually mild; mPAP >35-40 ('severe' / out of proportion to the lung disease) should prompt looking for a second cause and referral to a PH center. Cornerstone therapy is treating hypoxia (long-term O2), not PAH drugs.

BOARD TRAP

WRONG: attributing markedly elevated mPAP (e.g., 44 mmHg) entirely to COPD and starting PAH-specific vasodilators. Discriminator: Group 3 PH is typically modest — severe/out-of-proportion PH or a high PAWP signals an added cause (left heart, CTEPH); PAH drugs in Group 3 worsen V/Q matching and hypoxemia.